Supporting Information

Quantifying Indirect Contribution from Precursors to Human Body Burden of Legacy PFASs Based on Paired Blood and One-Week Duplicate Diet

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Chemicals and reagents

PFAC-MXB (a mixed standard solution containing 14 legacy PFAAs (C_{4-13} -PFCAs) and $C_{4,6,8,10}$-PFSAs), the mass labeled internal standard MPFAC-MXA (a mixture internal standards for corresponding PFCAs and PFSAs), and other single native and mass-labeled internal standards including four PFOS precursors (PFOSA, N-substituted PFOSA derivatives including FOSAA, N-EtFOSA, and N-EtFOSAA), four PFCA precursors (6:2 diPAP, 8:2 diPAP, 6:2 FTS, and 8:2 FTSs) and other two novel PFASs (6:2 Cl-PFESA and 8:2 Cl-PFESAs) were purchased from Wellington Laboratories (Ontario, Canada). The mixed standards of the isomers of PFOS, PFHxS and PFOA (*br*-PFOSK, *br*-PFHxSK and *T*-PFOA) were also obtained from Wellington Laboratories (Ontario, Canada).

HPLC-grade methanol and acetonitrile were purchased from Fisher (MA, USA). Sodium hydroxide, tetrabutylammonium hydrogen sulfate, methyl tert-butyl ether and potassium persulfate were purchased from Jiangtian Technology (Tianjin, China). Ammonium hydroxide and ammonium acetate were obtained from Sigma-Aldrich (MO, USA).

Participants and sample collection

All the subjects provided written informed consent and agreed to the use of their blood and urine samples. Besides the basic information such as age, gender, weight, and health condition. To reduce the influence of demographic characteristics on the results, the volunteers were selectively recruited to have similar characteristics, such as education level; age (20-25); non-smoker; non-alcohol drinker. A 30-day food recipe and questionnaire for each subject was recorded, which showed that their food recipes didn't change much among weeks and each volunteer has fixed drinking water sources. Through such investigation, it was believed that the 7-day dietary sampling can represent their intake over a period of time (at least one month). Thus, each subject was required to provide his (or her) three meals each day for a continuous week (7 day) in October, 2019. One of the most important rules for food sample collection was to collect what the volunteers actually ate. As a consequence, most of the food samples were cooked while seldomly did we receive uncooked food (such as salad). The samples of each day were transported back to the laboratory. After weighing, food was divided into three kinds of subsamples based on the ingredients: starch originated food (SOF, including rice and flour-based products), plant originated food (POF, including a variety of vegetables and soy products), and animal originated food (AOF, including meat, eggs, fish and seafood). Thus, there were three food subsamples per person per day, and there was a total of 420 food samples for the 20 subjects for 7 days. Simultaneously, drinking water samples (n=20) and indoor dust (n=20) were collected. For drinking water samples, as volunteers reported to have fixed sources, the samples were provided by each volunteer from corresponding source. As to the dust samples, indoor dust was mainly obtained from table, top of the closet, bookshelf, bed and

holder of bed. At the last day of sampling, whole blood (n=20), and morning urine (n=20) were collected from each subject. Intake of food were confiscated 8 hours before the blood and urine sampling while drinking water was allowed. All the samples were stored at -20 °C before analysis.

Sample pretreatment

1. Food samples:

After frozen-dried, classified or wet-weight based pooled homogenized food samples (15 g) were weighed and spiked with mass-labeled internal standards. Approximately 6 mL 400 mM NaOH was added into samples and stored at 4 °C overnight, after which 4 mL 500 mM tetrabutylammonium hydrogen sulfate (TBAH) solution was added, followed with 6 mL 250 mM sodium carbonate buffer and 6 mL methyl tert-butyl ether (MTBE). Then the samples were vortexed, shaken at 250 rpm for 30 min and ultra-sonicated for 30 min. Afterwards, samples were centrifuged at 10,000 rpm for 15 min and organic phases were transferred to new polypropylene tubes. The extraction procedures described before were performed for another two times with 5 mL MTBE added each time. The combined organic extracts were dried under gentle nitrogen flow at 40 °C and reconstituted in 50 mL Milli Q water with 0.5 mL methanol. Then the solvents were loaded onto WAX SPE cartridges (500 mg/6 mL) after cartridges being pre-conditioned with 8 mL 0.5% ammonium hydroxide in methanol, 6 mL methanol and 6 mL Milli Q water. Targeted compounds were eluted with 5 mL methanol and 5 mL 0.5% ammonium hydroxide in methanol after the column was washed with 6 mL 25 mM ammonium acetate and 4 mL Milli Q water. The final extracts were dried under gentle nitrogen flow and reconstitute in 200 μL methanol, after which the extracts were centrifugated at 13,000 rpm at 4 °C for 15 min and 150 μL supernatants were transferred to autosampler vials.

2. Drinking water samples:

Drinking water samples (500 mL) were spiked with mass-labeled internal standards and loaded onto WAX SPE cartridges (500 mg/6 mL) after the cartridge were pre-conditioned with 8 mL 0.5% ammonium hydroxide in methanol, 6 mL methanol and 6 mL Milli Q water. Then the targeted compounds were eluted with 5 mL methanol and 5 mL 0.5% ammonium hydroxide in methanol. The extracts were dried under gentle nitrogen flow at 40 °C and reconstituted with 600 μL methanol, after which the extracts were centrifugated at 13,000 rpm at 4 °C for 15 min and 500 μL supernatants were transferred to autosampler vials.

3. Dust samples:

About 100 mg dust samples were weighed and spiked with mass-labeled internal standards, then 4 mL of methanol was added. Afterwards, samples were shaken at 220 rpm for 30 min and ultra-sonicated for 30 min. The mixtures were centrifuged at 10,000 rpm for 15 min, then the supernatants were transferred to new polypropylene tubes. Abovementioned procedures were triplicate performed, and 50 mg PestiCarb Sorbents were added to the combined extracts. The mixtures were centrifuged at 10,000 rpm for 15 min and the supernatants were transferred to new tubes. Finally, the extracts were dried under gentle nitrogen flow at 40 °C and reconstituted with 300 μL methanol, after which the extracts were centrifugated at 13,000 rpm at 4 °C for 15 min and 200 μL supernatants were transferred to autosampler vials.

4.Whole blood samples:

Precise 1 mL of whole blood samples were spiked with mass-labeled internal standards followed with 5 mL acetonitrile/MTBE (1:1, v:v) solvent. Then the mixtures

were vortexed, shaken at 250 rpm for 30 min and ultra-sonicated for 30 min. The extraction procedures were conducted another two times, and the combined extracts were reduced to around 3 mL under gentle nitrogen flow. Then the extracts were loaded onto PestiCarb cartridges (250 mg/6 mL) after pre-conditioned with 5 mL acetonitrile. The eludes were dried under nitrogen flow at 40 °C and reconstituted with 200 μL methanol, after which the extracts were centrifugated at 13,000 rpm at 4 °C for 15 min and 150 μL supernatants were transferred to autosampler vials.

5. Urine samples:

Spiked 50 mL urine samples were loaded onto WAX SPE cartridges (150 mg/6 mL) after cartridges were pre-conditioned with 3 mL methanol and 2.5 mL 0.1 M formic acid. Then the cartridges were wash with 3 mL 0.1 M formic acid, followed with 5 mL 0.1 M formic acid/methanol (7:3, v:v) solution and 1 mL 1% ammonium hydroxide. After which targeted compounds were eluded with 4 mL methanol and 4 mL 0.5% ammonium hydroxide in methanol. The elutes were evaporated under gentle nitrogen flow at 40 °C and reconstituted with 200 μL mixture of methanol and Milli-Q water (1:1, v:v), after which the extracts were centrifugated at 13,000 rpm at 4 °C for 15 min and 150 μL supernatants were transferred to autosampler vials.

6. Total oxidable precursors (TOP) assay of poled food samples

After ion-pair extraction described above, the organic phases were concentrated to approximately 0.4 mL under nitrogen flow and reconstituted in 6 mL methanol. The extracts were loaded onto PestiCarb cartridges after pre-conditioned with 5 mL methanol. The colorless elutes were then dried under nitrogen flow (1 mL of Milli-Q water was added

before concentration to avoid the loss of volatile fractions) and reconstituted in 50 mL Milli-Q water, after which approximately 180 mM sodium hydroxide and 60 mM potassium persulfate were added. Then the extracts were thermalized at 85 °C for 6 h in a temperature-controlled water bath, leading to oxidable precursor being transformed corresponding perfluoroalkyl acids. After oxidation and cooling down, the pH of the solutions was measured to ensure the pH was greater than 12.0, then adjusted to 7.0 using concentrated hydrochloric acid (approximately 6 M). Then the solutions were loaded onto WAX SPE cartridges (500 mg/6 mL) after cartridges being pre-conditioned with 8 mL 0.5% ammonium hydroxide in methanol, 6 mL methanol and 6 mL Milli Q water. The cartridges were washed with 6 mL Milli-Q water before eluted with 8 mL 0.5% ammonium hydroxide in methanol. The extracts were dried under gentle nitrogen flow at 40 °C and reconstituted in 800 μL methanol, after which the solutions were centrifugated at 13,000 rpm at 4 °C for 15 min and 500 μL supernatants were transferred to autosampler vials.

The total amount of precursors was calculated based on the concentration difference of PFAAs before and after TOP assay.[1, 2]

$$C_{\mathrm{PreFOA}} = C_{\Delta\mathrm{PFNA}} \times 1.3 + \frac{C_{\Delta\mathrm{PFHpA}}}{1.2} \quad \text{(S1)}$$

$$C_{\mathrm{PreFOS}} = C_{\Delta\mathrm{PFOA}} - C_{\mathrm{PreFOA}} \quad \text{(S2)}$$

$$C_{\mathrm{PreFHxA}} = C_{\Delta\mathrm{PFHpA}} \times 1.3 + \frac{C_{\Delta\mathrm{PFPeA}}}{1.3} \quad \text{(S3)}$$

$$C_{\mathrm{PreFHxS}} = C_{\Delta\mathrm{PFHxA}} - C_{\mathrm{PreFHxA}} \quad \text{(S4)}$$

C_{PreFOA}, C_{PreFOS}, C_{PreFHxA} and C_{PreFHxS} are the concentration of PFOA, PFOS, perfluorohexanoic acid (PFHxA) and PFHxS precursors (ng/g), respectively. $C_{\Delta\mathrm{PFNA}}$, $C_{\Delta\mathrm{PFOA}}$, $C_{\Delta\mathrm{PFHpA}}$, $C_{\Delta\mathrm{PFHxA}}$, and $C_{\Delta\mathrm{PFPeA}}$ are the increased concentration of perfluorononanoic

acid (PFNA), PFOA, perfluoroheptanoic acid (PFHpA), PFHxA, and perfluoroheptanoic acid (PFPeA) (ng/g), respectively.

Calculation of EDI

The total EDI was calculated as follow:

$$\mathrm{EDI_{total}} = \mathrm{EDI_{diet}} + \mathrm{EDI_{water}} + \mathrm{EDI_{dust}} + \mathrm{EDI_{air}} \quad \text{(S5)}$$

The EDIs with subscripts of diet, water, dust and air were calculated using the followings equation, respectively:

$$\mathrm{EDI_{diet}} = \frac{C_{\mathrm{SOF}} \times m_{\mathrm{SOF}} + C_{\mathrm{AOF}} \times m_{\mathrm{AOF}} + C_{\mathrm{POF}} \times m_{\mathrm{POF}}}{\mathrm{BW}} \quad \text{(S6)}$$

$$\mathrm{EDI_{water}} = \frac{C_{\mathrm{water}} \times V_{\mathrm{water}}}{\mathrm{BW}} \quad \text{(S7)}$$

$$\mathrm{EDI_{dust}} = \frac{C_{\mathrm{dust}} \times m_{\mathrm{dust}}}{\mathrm{BW}} \quad \text{(S8)}$$

$$\mathrm{EDI_{air}} = \frac{C_{\mathrm{air}} \times V_{\mathrm{air}}}{\mathrm{BW}} \quad \text{(S9)}$$

where, C_i is the mean concentration of a specific PFAS in matrix i (SOF, AOF, POF, drinking water, and dust), m or V is the daily intake mass or volume of a matrix, respectively, BW is the body weight of the participant. The $\mathrm{EDI_{air}}$ was calculated based on a previous study of our group, which was also conducted in Tianjin for local residents' home.[3]

Instrumental analysis

Analysis of 24 kinds of PFASs was performed using a Waters UPLC system coupled with a Waters Xevo TQS tandem mass spectrometry (MS/MS) operated in negative electrospray ionization (ESI) mode. All compounds were separated using a Waters UPLC BEH C18 column (1.7 μm; 2.1 mm×50 mm) except for PFOA, PFOS and PFHxS isomers. The column was maintained at 55 °C. Ten μL aliquot of sample was injected and analyzed by the following gradient method at 0.3 mL/min using 2 mM ammonium acetate in Milli Q water (mobile phase A) and 2 mM ammonium acetate in methanol (mobile phase B):

(1) for PFAAs and FTSs: the initial solvent composition was 75 % A and 25 % B. Then it increased to 85 % B by 5 min, continued to increase to 100% by 5.1 min, and hold until 7 min, then returned to initial conditions at 9 min.

(2) for PFOSA, FOSAA, N-EtFOSA, N-EtFOSAA, Cl-PFESAs and diPAPs: the initial solvent composition was 80 % A and 20 % B. Then it increased to 95 % B by 3.5 min, hold until 4 min, then returned to initial conditions at 4.1 min, and hold to 6 min.

(3) isomers of PFOA, PFOS and PFHxS were separated by the Ascentis Express column (100×2.1 mm, 2.7 μm particle size; Supelco, PA, USA). The column was maintained at 38 °C. The flow rate of the UPLC was 0.25 mL/min and followed the gradient: initial composition was 10% B (methanol) and 90 % A (3 mM formic acid in water, adjusted to pH 4.0 with ammonium hydroxide), held for 5 min, increased to 30 % B by 5 min, ramped to 70 % B by 16 min, and finally 100 % B by 16.5 min, held until 17.5 min, and back to initial conditions 10% B by 20 min, and the column equilibrated for a further 3 min.

Table S1. Nomenclatures, chemical structures and isomeric compositions of *br*-PFOSK, *T*-PFOA and *br*-PFHxSK.

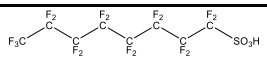
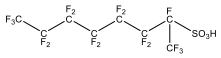
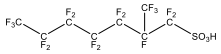
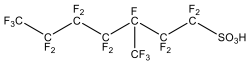
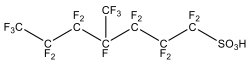
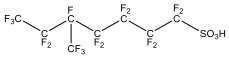
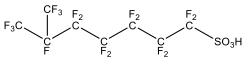
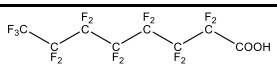
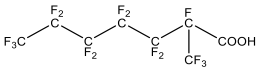
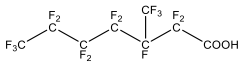
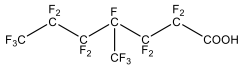
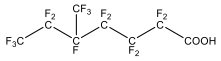
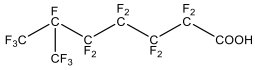
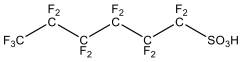
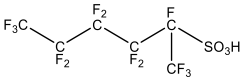
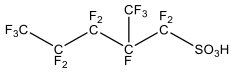
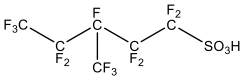
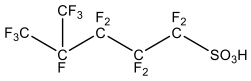
Nomenclature	Chemical structure	Composition (%)
n-PFOS		78.8
1m-PFOS		1.2
2m-PFOS		1.4
3m-PFOS		1.9
4m-PFOS		2.2
5m-PFOS		4.5
iso-PFOS		10.0
n-PFOA		79.0
2m-PFOA		0.5
3m-PFOA		3.0
4m-PFOA		4.0
5m-PFOA		4.5
iso-PFOA		9.0
n-PFHxS		81.1
1m- PFHxS		2.9
2m- PFHxS		1.4
3m- PFHxS		5.0
iso- PFHxS		8.9

Table S2. UPLC-MS/MS performance parameters.

Chemicals	Quantification transitions	Cone voltage (V)	Collision voltage(V)	Internal standard
PFBA	213→169	18	8	^{13}C-PFBA
PFPeA	263→219	19	8	^{13}C-PFHxA
PFHxA	313→269	12	9	^{13}C-PFHxA
PFHpA	363→319	16	10	^{13}C-PFOA
n-PFOA	413→369	15	10	^{13}C-PFOA
3m, iso-PFOA	413→169	15	17	^{13}C-PFOA
4m-PFOA	413→119	15	25	^{13}C-PFOA
5m-PFOA	413→219	15	15	^{13}C-PFOA
PFNA	463→419	15	11	^{13}C-PFNA
PFDA	513→469	15	11	^{13}C-PFDA
PFUnA	563→519	15	11	^{13}C-PFUdA
PFDoA	613→569	18	12	^{13}C-PFDoA
PFTrDA	663→619	16	22	^{13}C-PFDoA
PFBS	299→99	40	25	^{13}C-PFBS
n-PFHxS	399→119	40	30	^{18}O-PFHxS
1m-PFHxS	399→161	40	25	^{18}O-PFHxS
	399→319	40	25	
2m-PFHxS	399→130	40	35	^{18}O-PFHxS
	399→269	40	35	
3m-PFHxS	399→261	40	35	^{18}O-PFHxS
	399→219	40	35	
	399→180	40	25	
4m-PFHxS	399→169	40	35	^{18}O-PFHxS
	399→230	40	25	
n-PFOS	499→80	45	40	^{13}C-PFOS
1m-PFOS	499→419	45	25	^{13}C-PFOS
(3+5)m-PFOS	499→130	45	40	^{13}C-PFOS
4m-PFOS	499→330	45	30	^{13}C-PFOS
PFDS	599→80	50	40	^{13}C-PFDS
PFOSA	498→78	10	30	^{13}C-PFOSA
FOSAA	556→498	20	30	^{13}C-PFOSA
N-EtFOSA	526→169	58	26	^{2}H-NEtFOSA
N-EtFOSAA	584→419	18	20	^{2}H-NEtFOSA
6:2 diPAP	789→97	11	32	^{13}C-6:2 diPAP
8:2 diPAP	989→97	66	28	^{13}C-6:2 diPAP
6:2 FTS	427→407	54	26	^{13}C-8:2 FTS
8:2 FTS	527→507	8	22	^{13}C-8:2 FTS
6:2 Cl-PFESA	530→82.8	40	25	^{13}C-PFOS
8:2 Cl-PFESA	630.7→82.8	45	20	^{13}C-PFOS

Table S3. Limit of detection (LOD) and quantifications (LOQ) and recoveries (%, ± std) for PFASs.

Compounds	Diet			Drinking water			Dust			Blood			Urine		
	LOD (pg/g)	LOQ (pg/g)	Recovery (%)	LOD (pg/L)	LOQ (pg/L)	Recovery (%)	LOD (pg/g)	LOQ (pg/g)	Recovery (%)	LOD (pg/mL)	LOQ (pg/mL)	Recovery (%)	LOD (pg/L)	LOQ (pg/L)	Recovery (%)
PFBA	5	15	72±5	31	102	70±10	29	96	87±7	45	149	74±7	-	-	-
PFPeA	5	17	70±6	23	76	82±10	20	68	93±7	35	117	78±6	-	-	-
PFHxA	4	12	89±7	9	30	117±9	15	49	103±7	29	96	90±4	44	147	90±11
PFHpA	3	9	104±7	0	0.01	94±8	16	52	91±8	12	40	83±5	33	110	84±6
n-PFOA	2	6	94±6	0	0.01	106±13	21	71	98±6	4	13	92±7	127	422	95±7
3m-PFOA	0.3	1	92±7	-	-	-	-	-	-	1	2	90±5	98	327	93±5
4m-PFOA	1	3	97±5	-	-	-	-	-	-	1	2	93±7	12	39	89±11
5m-PFOA	1	2	97±6	-	-	-	-	-	-	0.2	0.59	92±5	13	43	90±8
iso-PFOA	1	4	94±7	-	-	-	-	-	-	1	3.57	95±6	10	34	93±8
PFNA	2	8	108±6	0	0.01	89±10	18	59	93±8	5	17	94±1	31	103	92±9
PFDA	3	9	128±3	4	14	117±10	20	68	87±10	5	16	88±3	26	86	97±9
PFUnA	2	8	96±7	0	0.00	96±6	17	55	92±6	5	16	87±10	35	117	83±6
PFDoA	3	9	84±6	0	0.00	87±3	22	73	101±3	9	29	104±8	39	129	86±4
PFTrDA	5	17	73±6	0	0.01	81±8	30	101	88±7	23	77	91±7	22	74	77±12
PFBS	15	49	72±4	0	0.02	107±9	28	92	85±10	22	74	80±6	-	-	-
n-PFHxS	2	8	99±7	0	0.04	112±7	26	86	89±4	16	54	103±9	44	145	88±7
1m-PFHxS	17	58	95±7	-	-	-	-	-	-	0.3	1	105±7	26	86	81±7
2m-PFHxS	2	6	94±6	-	-	-	-	-	-	0.3	1	101±7	15	49	82±8
3m-PFHxS	1	4	93±7	-	-	-	-	-	-	0.3	1	103±8	16	52	86±12
iso-PFHxS	3	11	99±5	-	-	-	-	-	-	0.3	1	107±10	12	41	89±10
n-PFOS	9	29	112±5	0	0.00	95±10	37	124	98±7	31	103	103±8	178	593	90±9
1m-PFOS	1	2	108±7	-	-	-	-	-	-	0.2	0.69	105±6	22	73	86±6
3+5m-PFOS	1	4	114±8	-	-	-	-	-	-	0.6	2	99±9	59	197	94±9
4m-PFOS	1	3	110±6	-	-	-	-	-	-	1	3	97±7	52	174	85±7
iso-PFOS	1	3	108±4	-	-	-	-	-	-	1	4	98±6	50	167	92±6
PFDS	11	38	91±5	1	2	94±7	35	117	88±9	38	128	83±3	57	189	83±8
PFOSA	5	16	102±3	0	0.00	91±10	15	49	94±6	1	4	113±5	18	59	94±8
FOSAA	7	24	108±4	3	10	97±7	25	82	101±9	1	4	114±6	-	-	-
N-EtFOSA	24	81	105±4	14	48	94±7	36	121	94±6	13	42	111±6	-	-	-
N-EtFOSAA	23	77	120±5	11	37	104±4	35	117	92±6	13	42	129±2	-	-	-
6:2 diPAP	22	73	130±4	4	12	104±7	47	158	105±7	16	54	106±6	-	-	-
8:2 diPAP	32	108	113±5	6	21	94±5	67	224	91±7	24	79	118±2	-	-	-
6:2 FTS	14	48	80±5	9	29	91±8	30	101	97±7	22	73	81±8	19	64	83±6
8:2 FTS	18	59	118±5	13	43	88±7	53	177	88±3	28	92	88±7	23	75	74±4
6:2 Cl-PFESA	5	16	95±6	0	0.00	111±4	25	84	95±10	**9**	**29**	97±6	13	44	88±10
8:2 Cl-PFESA	6	21	91±8	6	19	102±2	31	102	92±10	**19**	**62**	95±8	28	93	81±8

Table S4. Concentration (ng/g) of PFASs in three kind of diet ingredients (SOF, POF and AOF) of the volunteers

	PFBS			PFHxS			PFOS			PFBA			PFPeA		
	SOF	POF	AOF	POF	POF	AOF	SOF	POF	AOF	SOF	POF	AOF	SOF	POF	AOF
1	0.00	0.00	0.00	0.03±0.01	0.03±0.01	0.22±0.03	0.12±0.01	0.11±0.03	0.71±0.13	0.00	0.03±0.01	0.05±0.02	0.12±0.01	0.06±0.02	0.08±0.02
2	0.00	0.00	0.00	0.04±0.02	0.04±0.02	0.26±0.05	0.13±0.01	0.31±0.02	0.40±0.11	0.00	0.04±0.02	0.04±0.01	0.14±0.02	0.07±0.02	0.16±0.01
3	0.00	0.10±0.07	0.00	0.02±0.01	0.02±0.01	0.29±0.06	0.26±0.01	0.28±0.04	3.00±0.27	0.03±0.01	0.02±0.01	0.03±0.02	0.13±0.00	0.05±0.03	0.08±0.03
4	0.00	0.00	0.00	0.00	0.00	0.48±0.07	0.37±0.02	0.25±0.03	1.70±0.09	0.00	0.00	0.00	0.16±0.02	0.10±0.01	0.15±0.02
5	0.00	0.13±0.02	0.00	0.00	0.00	0.25±0.04	0.48±0.01	0.27±0.02	0.51±0.12	0.04±0.02	0.00	0.04±0.01	0.16±0.01	0.07±0.03	0.11±0.02
6	0.00	0.00	0.00	0.00	0.00	0.36±0.09	0.22±0.01	0.26±0.04	1.29±0.08	0.00	0.00	0.05±0.03	0.00	0.00	0.25±0.04
7	0.00	0.11±0.05	0.09	0.03±0.01	0.03±0.01	0.52±0.13	0.20±0.03	0.21±0.05	2.15±0.37	0.00	0.03±0.01	0.00	0.10±0.01	0.05±0.02	0.09±0.03
8	0.00	0.00	0.00	0.03±0.01	0.03±0.01	0.26±0.04	0.13±0.02	0.08±0.02	0.59±0.08	0.00	0.03±0.01	0.00	0.10±0.00	0.08±0.03	0.13±0.02
9	0.00	0.00	0.00	0.00	0.00	0.23±0.06	0.28±0.01	0.17±0.03	0.42±0.06	0.02±0.01	0.00	0.05±0.01	0.13±0.01	0.09±0.02	0.08±0.03
10	0.00	0.00	0.00	0.00	0.00	0.22±0.05	0.12±0.01	0.15±0.02	0.38±0.07	0.00	0.00	0.05±0.01	0.15±0.02	0.06±0.03	0.00
11	0.00	0.00	0.00	0.04±0.01	0.04±0.01	0.14±0.02	0.21±0.02	0.68±0.09	0.38±0.16	0.00	0.04±0.01	0.00	0.00	0.00	0.06±0.02
12	0.00	0.00	0.00	0.04±0.03	0.04±0.03	0.33±0.07	0.14±0.02	0.14±0.03	0.40±0.07	0.00	0.04±0.03	0.00	0.12±0.02	0.00	0.74±0.01
13	0.00	0.00	0.00	0.00	0.00	0.10±0.01	0.14±0.01	0.16±0.02	0.48±0.03	0.00	0.00	0.03±0.02	0.02±0.01	0.09±0.03	0.09±0.03
14	0.00	0.07±0.02	0.00	0.03±0.01	0.03±0.01	0.10±0.06	0.14±0.00	0.16±0.01	0.55±0.10	0.00	0.03±0.01	0.04±0.01	0.03±0.00	0.10±0.03	0.10±0.01
15	0.00	0.05±0.03	0.00	0.00	0.00	0.08±0.04	0.14±0.00	0.15±0.02	0.51±0.09	0.00	0.00	0.00	0.04±0.01	0.10±0.03	0.10±0.00
16	0.00	0.08±0.02	0.00	0.00	0.00	0.12±0.02	0.14±0.00	0.16±0.01	0.53±0.09	0.00	0.00	0.00	0.02±0.00	0.10±0.02	0.10±0.02
17	0.00	0.05±0.01	0.00	0.00	0.00	0.08±0.03	0.14±0.01	0.17±0.02	0.65±0.07	0.00	0.00	0.03±0.01	0.04±0.00	0.10±0.01	0.10±0.01
18	0.00	0.05±0.02	0.00	0.03±0.01	0.03±0.01	0.09±0.03	0.14±0.00	0.16±0.03	0.63±0.06	0.00	0.03±0.01	0.00	0.03±0.01	0.10±0.02	0.10±0.01
19	0.00	0.00	0.00	0.03±0.02	0.03±0.02	0.08±0.03	0.14±0.01	0.16±0.00	0.59±0.07	0.02±0.01	0.03±0.02	0.02±0.01	0.03±0.01	0.10±0.02	0.10±0.01
20	0.00	0.07±0.01	0.00	0.00	0.00	0.10±0.02	0.14±0.00	0.16±0.01	0.55±0.13	0.00	0.00	0.04±0.01	0.03±0.01	0.10±0.01	0.10±0.02
Mean	0.00	0.04	0.00	0.02	0.02	0.22	0.19	0.21	0.82	0.01	0.02	0.02	0.08	0.07	0.14
Median	0.00	0.00	0.00	0.01	0.01	0.22	0.14	0.16	0.55	0.00	0.01	0.03	0.07	0.08	0.10
DF (%)	0	45	0	50	50	100	100	100	100	20	50	60	90	95	95

	PFHxA			PFHpA			PFOA			PFNA			PFDA		
	SOF	POF	AOF	SOF	POF	AOF	SOF	POF	AOF	SOF	POF	AOF	SOF	POF	AOF
1	0.15±0.01	0.15±0.04	0.19±0.05	0.28±0.03	0.26±0.07	0.19±0.10	0.57±0.06	1.24±0.11	1.51±0.17	0.20±0.02	0.19±0.02	0.48±0.11	0.03±0.00	0.05±0.02	0.07±0.03
2	0.19±0.00	0.36±0.04	0.29±0.04	0.26±0.02	0.21±0.11	0.21±0.05	0.27±0.03	0.48±0.06	0.48±0.06	0.68±0.09	0.00	1.41±0.02	0.06±0.00	0.06±0.01	0.14±0.01
3	0.22±0.02	0.20±0.05	0.42±0.13	0.12±0.03	0.23±0.05	0.26±0.07	0.63±0.04	1.32±0.09	1.83±0.04	0.23±0.03	0.42±0.06	0.54±0.06	0.05±0.01	0.05±0.02	0.16±0.03
4	0.14±0.01	0.15±0.02	0.39±0.07	0.15±0.01	0.14±0.07	0.26±0.06	1.19±0.08	2.24±0.17	2.98±0.32	0.33±0.03	0.28±0.04	0.69±0.05	0.03±0.01	0.03±0.01	0.15±0.04
5	0.22±0.02	0.17±0.04	0.19±0.05	0.40±0.02	0.15±0.06	0.18±0.04	0.86±0.05	1.79±0.08	1.94±0.06	0.19±0.03	0.15±0.02	0.29±0.05	0.03±0.00	0.02±0.00	0.04±0.02
6	0.21±0.02	0.16±0.03	0.33±0.11	0.16±0.02	0.08±0.03	0.21±0.07	0.17±0.02	0.28±0.06	0.64±0.04	0.20±0.05	0.13±0.02	0.39±0.03	0.02±0.00	0.03±0.01	0.04±0.03
7	0.12±0.01	0.13±0.06	0.35±0.08	0.13±0.01	0.07±0.02	0.09±0.06	0.17±0.03	0.26±0.05	0.51±0.07	0.12±0.02	0.10±0.01	0.33±0.04	0.03±0.00	0.00	0.12±0.05
8	0.13±0.01	0.18±0.03	0.27±0.03	0.19±0.01	0.20±0.06	0.14±0.05	0.14±0.02	0.38±0.03	0.54±0.04	0.23±0.02	0.21±0.03	0.27±0.05	0.05±0.01	0.05±0.02	0.04±0.01
9	0.16±0.02	0.11±0.02	0.14±0.12	0.40±0.06	0.11±0.04	0.16±0.03	0.20±0.02	0.35±0.07	0.47±0.08	0.15±0.03	0.75±0.11	0.20±0.06	0.10±0.02	0.08±0.01	0.04±0.02
10	0.12±0.02	0.11±0.04	0.12±0.03	0.11±0.00	0.08±0.04	0.13±0.03	0.20±0.01	0.28±0.04	0.40±0.05	0.28±0.03	0.19±0.02	0.30±0.03	0.00	0.00	0.03±0.01
11	0.10±0.02	0.09±0.06	0.12±0.06	0.07±0.03	0.13±0.02	0.08±0.01	0.12±0.01	0.22±0.02	0.30±0.04	0.18±0.03	0.25±0.01	0.49±0.04	0.00	0.10±0.02	0.05±0.04
12	0.12±0.01	0.14±0.03	0.18±0.11	0.12±0.03	0.20±0.04	0.14±0.06	0.15±0.02	0.23±0.06	0.33±0.08	0.62±0.06	0.34±0.05	0.18±0.06	0.13±0.01	0.02±0.00	0.03±0.02
13	0.09±0.01	0.15±0.04	0.19±0.04	0.06±0.02	0.04±0.01	0.03±0.01	0.20±0.00	0.29±0.01	0.62±0.11	0.07±0.01	0.09±0.00	0.21±0.03	0.00	0.05±0.01	0.09±0.03
14	0.09±0.02	0.16±0.03	0.20±0.06	0.05±0.01	0.06±0.02	0.04±0.03	0.20±0.00	0.33±0.05	0.61±0.07	0.07±0.00	0.09±0.02	0.20±0.05	0.00	0.06±0.01	0.08±0.02
15	0.09±0.01	0.15±0.03	0.20±0.08	0.05±0.01	0.06±0.04	0.03±0.02	0.20±0.02	0.29±0.07	0.62±0.09	0.07±0.00	0.09±0.01	0.20±0.03	0.00	0.06±0.01	0.08±0.00
16	0.09±0.01	0.16±0.04	0.20±0.13	0.06±0.02	0.06±0.02	0.04±0.02	0.20±0.01	0.32±0.02	0.61±0.15	0.07±0.02	0.08±0.00	0.21±0.06	0.00	0.04±0.00	0.07±0.02
17	0.09±0.02	0.17±0.04	0.21±0.04	0.05±0.01	0.05±0.01	0.03±0.02	0.21±0.01	0.36±0.07	0.59±0.09	0.07±0.01	0.09±0.03	0.19±0.02	0.00	0.06±0.03	0.07±0,01
18	0.09±0.01	0.16±0.03	0.21±0.03	0.06±0.01	0.05±0.03	0.03±0.02	0.20±0.00	0.33±0.05	0.59±0.06	0.07±0.01	0.09±0.02	0.19±0.07	0.00	0.05±0.01	0.07±0.01
19	0.09±0.01	0.16±0.04	0.21±0.05	0.06±0.00	0.06±0.01	0.03±0.01	0.20±0.00	0.32±0.04	0.60±0.05	0.07±0.01	0.09±0.00	0.19±0.04	0.00	0.05±0.00	0.08±0.02
20	0.09±0.01	0.16±0.03	0.20±0.08	0.06±0.00	0.06±0.01	0.04±0.02	0.20±0.00	0.33±0.04	0.61±0.05	0.07±0.00	0.09±0.03	0.20±0.06	0.00	0.06±0.00	0.08±0.00
Mean	0.13	0.16	0.23	0.14	0.11	0.12	0.31	0.58	0.84	0.20	0.18	0.36	0.03	0.05	0.08
Median	0.12	0.16	0.20	0.11	0.08	0.11	0.20	0.33	0.60	0.16	0.11	0.24	0.01	0.05	0.07
DF (%)	100	100	100	100	100	100	100	100	100	100	100	100	50	90	100

	PFUdA			PFDoA			PFOSA			6:2 Cl-PFESA		
	SOF	POF	AOF	SOF	POF	AOF	SOF	POF	AOF	SOF	POF	AOF
1	0.04±0.00	0.03±0.01	0.06±0.02	0.12±0.02	0.12±0.02	0.14±0.02	0.21±0.03	0.18±0.03	0.22±0.04	0.23±0.03	0.19±0.03	0.40±0.06
2	0.03±0.00	0.03±0.02	0.07±0.01	0.08±0.02	0.10±0.01	0.19±0.01	0.11±0.02	0.13±0.03	0.16±0.03	0.18±0.01	0.24±0.03	0.34±0.03
3	0.07±0.01	0.12±0.04	0.10±0.01	0.07±0.01	0.11±0.02	0.11±0.02	0.13±0.03	0.11±0.02	0.15±0.03	0.17±0.01	0.17±0.01	0.22±0.03
4	0.10±0.02	0.15±0.02	0.08±0.03	0.17±0.02	0.08±0.01	0.17±0.03	0.13±0.03	0.15±0.03	0.14±0.05	0.14±0.02	0.17±0.02	0.25±0.03
5	0.09±0.00	0.04±0.01	0.06±0.02	0.06±0.00	0.04±0.02	0.05±0.02	0.12±0.02	0.14±0.04	0.16±0.03	0.29±0.01	0.27±0.02	0.42±0.04
6	0.06±0.01	0.05±0.03	0.16±0.03	0.06±0.01	0.05±0.02	0.06±0.02	0.15±0.03	0.11±0.03	0.17±0.07	0.17±0.02	0.19±0.04	0.35±0.04
7	0.06±0.01	0.06±0.01	0.19±0.04	0.04±0.01	0.03±0.01	0.07±0.01	0.14±0.05	0.19±0.03	0.17±0.05	0.11±0.01	0.19±0.02	0.23±0.07
8	0.06±0.00	0.04±0.00	0.05±0.03	0.07±0.02	0.06±0.01	0.07±0.02	0.13±0.02	0.12±0.02	0.16±0.03	0.12±0.00	0.17±0.04	0.26±0.03
9	0.04±0.00	0.19±0.03	0.19±0.02	0.06±0.01	0.04±0.02	0.00	0.10±0.03	0.15±0.04	0.15±0.03	0.14±0.01	0.13±0.02	0.04±0.01
10	0.06±0.01	0.05±0.01	0.28±0.07	0.00	0.00	0.04±0.01	0.14±0.01	0.13±0.04	0.15±0.04	0.12±0.02	0.15±0.01	0.06±0.00
11	0.04±0.01	0.05±0.02	0.09±0.03	0.00	0.06±0.01	0.08±0.01	0.18±0.02	0.15±0.02	0.19±0.05	0.15±0.01	0.10±0.04	0.05±0.01
12	0.11±0.02	0.05±0.02	0.15±0.03	0.15±0.04	0.05±0.01	0.04±0.02	0.15±0.03	0.14±0.06	0.19±0.05	0.16±0.01	0.25±0.06	0.04±0.01
13	0.00	0.04±0.01	0.06±0.01	0.00	0.00	0.03±0.02	0.02±0.00	0.03±0.01	0.04±0.02	0.07±0.01	0.19±0.02	0.29±0.05
14	0.00	0.04±0.00	0.06±0.02	0.00	0.01±0.02	0.03±0.01	0.02±0.00	0.02±0.00	0.04±0.02	0.07±0.00	0.19±0.03	0.28±0.02
15	0.00	0.04±0.01	0.06±0.04	0.00	0.02±0.00	0.04±0.00	0.02±0.00	0.02±0.00	0.04±0.03	0.07±0.02	0.17±0.01	0.29±0.04
16	0.00	0.04±0.00	0.07±0.02	0.00	0.02±0.01	0.03±0.01	0.02±0.01	0.02±0.00	0.04±0.01	0.07±0.01	0.20±0.02	0.28±0.04
17	0.00	0.04±0.00	0.06±0.03	0.00	0.01±0.00	0.03±0.01	0.02±0.00	0.02±0.01	0.04±0.00	0.07±0.00	0.19±0.03	0.27±0.03
18	0.00	0.04±0.01	0.06±0.02	0.00	0.01±0.00	0.03±0.00	0.02±0.00	0.02±0.02	0.04±0.02	0.07±0.00	0.18±0.02	0.27±0.05
19	0.00	0.04±0.02	0.06±0.00	0.00	0.02±0.00	0.03±0.00	0.02±0.00	0.02±0.01	0.04±0.02	0.07±0.01	0.18±0.03	0.28±0.03
20	0.00	0.04±0.00	0.06±0.03	0.00	0.01±0.01	0.03±0.01	0.02±0.01	0.02±0.00	0.04±0.01	0.07±0.01	0.19±0.02	0.28±0.06
Mean	0.04	0.06	0.10	0.04	0.04	0.06	0.09	0.09	0.12	0.13	0.18	0.23
Median	0.04	0.04	0.07	0.02	0.03	0.04	0.12	0.12	0.15	0.12	0.19	0.27
DF (%)	60	100	100	50	90	95	100	100	100	100	100	100

	8:2 Cl-PFESA			ΣPFASs		
	SOF	POF	AOF	SOF	POF	AOF
1	0.00	0.02±0.01	0.08±0.02	2.14±0.20	2.72±0.43	4.40±0.82
2	0.06±0.03	0.04±0.02	0.08±0.03	2.31±0.22	2.18±0.51	4.23±0.79
3	0.00	0.04±0.02	0.13±0.03	2.17±019	3.34±0.62	7.33±1.58
4	0.00	0.07±0.02	0.00	3.14±0.27	3.90±0.57	7.43±1.69
5	0.00	0.00	0.00	3.22±0.35	3.37±0.43	4.25±1.01
6	0.06±0.03	0.06±0.01	0.17±0.05	1.54±0.11	1.47±0.27	4.47±0.68
7	0.00	0.00	0.00	1.32±0.12	1.64±0.21	4.92±0.71
8	0.00	0.04±0.02	0.03±0.02	1.41±0.14	1.71±0.28	2.81±0.44
9	0.00	0.00	0.00	1.90±0.15	2.27±0.30	2.14±0.37
10	0.00	0.07±0.02	0.00	1.40±0.13	1.27±0.27	2.10±0.41
11	0.06±0.04	0.06±0.03	0.09±0.03	1.17±0.11	2.02±0.46	2.06±0.35
12	0.00	0.00	0.00	2.01±0.15	1.66±0.37	2.72±0.48
13	0.00	0.12±0.04	0.13±0.04	0.68±0.07	1.28±0.11	2.41±0.55
14	0.00	0.11±0.03	0.12±0.03	0.68±0.04	1.47±0.18	2.46±0.48
15	0.00	0.11±0.03	0.13±0.03	0.68±0.06	1.34±0.13	2.37±0.51
16	0.00	0.12±0.03	0.12±0.03	0.68±0.06	1.43±0.13	2.42±0.43
17	0.00	0.11±0.01	0.12±0.04	0.69±0.05	1.46±0.12	2.47±0.29
18	0.00	0.11±0.03	0.12±0.03	0.68±0.04	1.44±0.12	2.43±0.46
19	0.00	0.11±0.04	0.12±0.03	0.70±0.06	1.38±0.11	2.43±0.55
20	0.00	0.11±0.02	0.12±0.04	0.68±0.07	1.43±0.15	2.45±0.57
Mean	0.01	0.06	0.08	1.46	1.94	3.41
Median	0.00	0.06	0.10	1.17	1.54	2.76
DF (%)	15	80	70	-	-	-

Table S5. Concentration (ng/L) of PFASs in drinking water samples

	PFBS	PFHxS	PFOS	PFBA	PFHxA	PFHpA	PFOA	PFNA	PFDA	PFUdA	6:2 Cl-PFESA	ΣPFASs
1	0.06	0.14	0.49	0.04	0.43	0.05	0.78	0.08	0.09	0.09	0.56	2.81
2	0.06	0.14	0.48	0.00	0.41	0.04	0.83	0.08	0.10	0.09	0.50	2.73
3	0.06	0.14	0.49	0.04	0.44	0.05	0.74	0.07	0.09	0.09	0.63	2.84
4	0.06	0.14	0.50	0.00	0.50	0.05	0.60	0.07	0.08	0.09	0.82	2.91
5	0.06	0.14	0.50	0.04	0.50	0.05	0.60	0.07	0.08	0.09	0.82	2.95
6	0.06	0.14	0.49	0.00	0.42	0.04	0.80	0.08	0.10	0.09	0.53	2.75
7	0.06	0.14	0.50	0.00	0.48	0.05	0.64	0.07	0.09	0.09	0.76	2.88
8	0.06	0.14	0.49	0.04	0.43	0.05	0.78	0.08	0.09	0.09	0.56	2.81
9	0.06	0.14	0.49	0.04	0.45	0.05	0.71	0.07	0.09	0.09	0.66	2.85
10	0.06	0.14	0.49	0.04	0.43	0.05	0.78	0.08	0.09	0.09	0.56	2.81
11	0.06	0.14	0.51	0.04	0.52	0.06	0.55	0.06	0.08	0.09	0.89	3
12	0.06	0.14	0.51	0.03	0.54	0.06	0.51	0.06	0.08	0.09	0.95	3.03
13	0.05	0.16	0.42	0.04	0.75	0.06	0.93	0.14	0.10	0.07	0.45	3.17
14	0.05	0.16	0.42	0.04	0.75	0.06	0.93	0.14	0.10	0.07	0.45	3.17
15	0.05	0.16	0.42	0.04	0.75	0.06	0.93	0.14	0.10	0.07	0.45	3.17
16	0.05	0.16	0.42	0.04	0.75	0.06	0.93	0.14	0.10	0.07	0.45	3.17
17	0.05	0.16	0.42	0.04	0.75	0.06	0.93	0.14	0.10	0.07	0.45	3.17
18	0.05	0.16	0.42	0.04	0.75	0.06	0.93	0.14	0.10	0.07	0.45	3.17
19	0.05	0.16	0.42	0.04	0.75	0.06	0.93	0.14	0.10	0.07	0.45	3.17
20	0.05	0.16	0.42	0.04	0.75	0.06	0.93	0.14	0.10	0.07	0.45	3.17
Mean	0.06	0.15	0.46	0.03	0.58	0.05	0.79	0.10	0.09	0.08	0.59	
Median	0.06	0.14	0.49	0.04	0.51	0.06	0.79	0.08	0.10	0.09	0.55	
DF (%)	100	100	100	80	100	100	100	100	100	100	100	

Table S6. Concentrations of PFASs (ng/L) in drinking water in China

Year	Reference	source	PFHxS	PFOS	PFHxA	PFOA	PFNA	6:2 Cl-PFESA
2002-2006	[4]	tap water	-	2.1	-	4.7	-	-
2011	[5]	tap water	-	0.35	-	0.30	-	-
2014-2015	[6]	drinking water treatment plant	-	5.0	0.7	4.8	0.8	-
2015-2017	[7]	tap water	6.3	7.7	4.1	9.9	-	-
2017	[8]	tap water	0.90	1.22	-	1.81	0.31	-
2019	this study	purified or bottled water	0.15	0.46	0.58	0.79	0.10	0.59

Table S7. Concentration (ng/g) of PFASs in the indoor dust.

	PFBS	PFHxS	PFOS	PFBA	PFPeA	PFHxA	PFHpA	PFOA	PFNA	PFDA	PFUdA	PFDoA	PFOSA	N-EtFOSA	6:2 diPAP	8:2 diPAP	6:2 Cl-PFESA	8:2 Cl-PFESA	ΣPFASs
1	0.39	0.71	3.82	0.14	0.27	4.43	1.41	4.09	1.46	1.27	1.04	0.75	0.07	0.00	3.78	1.75	2.73	1.57	29.71
2	0.26	0.22	2.54	0.00	0.39	3.74	2.31	5.20	0.64	0.88	0.33	0.35	0.22	0.35	1.17	2.87	0.89	0.36	22.73
3	0.23	0.22	2.54	0.10	0.26	3.74	1.78	3.58	0.59	1.04	0.29	0.40	0.08	0.00	2.52	1.53	0.84	0.43	20.17
4	0.48	0.49	3.33	0.13	0.27	3.17	2.07	3.65	0.95	2.02	0.38	0.42	0.08	0.00	2.40	0.00	0.88	0.53	21.22
5	0.00	0.52	2.89	0.00	0.18	2.85	1.41	4.69	0.69	1.08	0.28	0.29	0.06	0.00	3.74	0.00	0.69	0.29	19.68
6	0.39	0.71	3.82	0.14	0.27	4.43	1.41	5.09	1.46	1.27	1.04	0.75	0.11	0.00	3.78	1.75	2.73	1.57	30.75
7	0.30	0.16	2.10	0.00	0.09	2.16	0.65	8.89	0.37	1.59	0.52	0.26	0.08	0.00	4.07	4.67	0.91	0.45	27.27
8	0.18	0.20	0.93	0.08	0.14	2.58	0.49	2.72	0.30	1.06	0.13	0.31	0.13	0.00	1.67	0.00	0.81	0.29	12.02
9	0.23	0.29	1.74	0.10	0.13	2.96	0.51	3.89	0.35	1.49	0.30	0.29	0.06	0.00	2.77	0.00	0.74	0.32	16.18
10	0.23	0.29	1.74	0.10	0.13	2.96	0.51	3.89	0.35	1.49	0.30	0.29	0.06	0.00	2.77	0.00	0.74	0.38	16.23
11	0.20	0.21	2.57	0.00	0.14	3.65	1.79	4.84	0.29	0.91	0.23	0.25	0.00	0.00	1.57	0.00	1.28	0.29	18.22
12	0.34	0.21	2.42	0.15	0.17	3.12	1.18	6.18	0.27	0.88	0.22	0.27	0.00	0.48	2.02	3.82	1.49	0.32	23.54
13	0.12	0.15	0.20	0.00	0.21	3.92	0.56	2.24	0.81	0.75	0.56	0.77	0.10	0.18	0.36	0.84	0.42	0.24	12.43
14	0.12	0.15	0.20	0.00	0.21	3.92	0.56	2.24	0.81	0.75	0.56	0.77	0.10	0.18	0.36	0.84	0.42	0.24	12.43
15	0.16	0.18	0.16	0.10	0.27	2.58	0.56	2.55	0.91	0.84	0.29	0.79	0.05	0.00	0.34	0.72	0.25	0.22	10.98
16	0.16	0.18	0.16	0.10	0.27	2.58	0.52	2.55	0.91	0.84	0.29	0.79	0.05	0.00	0.34	0.72	0.25	0.22	10.94
17	0.16	0.13	0.24	0.10	0.27	2.58	0.52	2.55	0.91	0.84	0.29	0.79	0.05	0.00	0.34	0.72	0.25	0.22	10.97
18	0.16	0.13	0.24	0.10	0.27	2.58	0.52	2.55	0.91	0.84	0.29	0.79	0.05	0.00	0.34	0.72	0.25	0.22	10.97
19	0.13	0.13	0.24	0.00	0.15	0.32	0.27	0.93	0.25	0.59	0.12	0.37	0.00	0.00	0.30	1.04	0.31	0.20	5.35
20	0.13	0.13	0.24	0.00	0.15	0.32	0.27	0.93	0.25	0.59	0.12	0.37	0.00	0.00	0.30	1.04	0.31	0.20	5.35
Mean	0.22	0.27	1.61	0.07	0.21	2.93	0.97	3.66	0.67	1.05	0.38	0.50	0.07	0.06	1.75	1.15	0.86	0.43	29.71
Mediann	0.19	0.20	1.74	0.10	0.21	2.96	0.56	3.61	0.67	0.90	0.29	0.38	0.06	0.00	1.62	0.78	0.74	0.29	
DF (%)	100	100	100	100	100	100	100	100	100	100	100	100	85	20	100	75	100	100	

Table S8. EDIs (ng/kg/day) of the four main PFASs on an individual body weight basis.

	PFOA	PFOS	PFHxS	6:2 Cl-PFESA
SOF				
Mean	1.66	1.02	0.33	0.67
Median	1.15	0.79	0.31	0.61
Min	0.51	0.50	0.02	0.31
Max	5.63	2.56	1.40	1.57
POF				
Mean	2.42	0.95	0.33	0.80
Median	1.32	0.70	0.34	0.69
Min	0.86	0.31	0.00	0.52
Max	9.23	3.65	0.90	1.28
AOF				
Mean	1.54	1.39	0.40	0.39
Median	0.76	0.81	0.37	0.32
Min	0.37	0.33	0.08	0.00
Max	7.06	4.31	1.14	0.97
Total Diet				
Mean	5.62	3.36	1.07	1.86
Median	3.38	2.53	1.13	1.74
Min	2.36	1.66	0.10	1.13
Max	21.93	6.79	2.64	3.56
Drinking water				
Mean	0.02	0.01	0.003	0.01
Median	0.02	0.01	0.004	0.01
Min	0.01	0.01	0.002	0.01
Max	0.02	0.02	0.004	0.03
Dust				
Mean	0.05	0.001	0	0
Median	0.04	0.001	0	0
Min	0.01	0.000	0	0
Max	0.18	0.003	0	0.002
Total Daily Intake				
Mean	5.68	3.37	1.07	1.87
Median	3.45	2.55	1.14	1.75
Min	2.50	1.66	0.11	1.16
Max	21.9	6.80	2.65	3.58

Table S9. Concentrations (ng/mL) of PFASs in the whole blood of the volunteers.

	PFBS	PFHxS	PFOS	PFBA	PFHxA	PFHpA	PFOA	PFNA	PFDA	PFUdA	PFOSA	6:2 diPAP	6:2 Cl-PFESA	8:2 Cl-PFESA	ΣPFASs
1	0.09	1.03	1.31	0.00	0.23	0.09	1.72	0.21	0.14	0.11	0.10	0.00	2.90	0.17	8.1
2	0.09	0.90	1.52	0.20	0.26	0.06	0.99	0.26	0.06	0.08	0.07	0.00	2.01	0.22	6.72
3	0.09	0.88	0.95	0.00	0.27	0.06	1.55	0.14	0.06	0.10	0.01	0.00	0.93	0.11	5.15
4	0.09	0.76	1.17	0.00	0.32	0.06	2.51	0.14	0.07	0.08	0.01	0.00	1.02	0.00	6.23
5	0.09	1.00	2.02	0.00	0.25	0.07	1.47	0.11	0.06	0.09	0.02	0.00	3.19	0.00	8.37
6	0.09	0.90	0.64	0.26	0.23	0.06	0.84	0.09	0.05	0.05	0.01	0.00	0.87	0.19	4.28
7	0.08	0.74	1.15	0.00	0.19	0.07	1.41	0.09	0.06	0.12	0.03	0.00	1.21	0.00	5.15
8	0.09	0.82	0.72	0.00	0.21	0.06	0.98	0.11	0.07	0.07	0.03	0.00	0.48	0.00	3.64
9	0.08	0.66	0.85	0.00	0.22	0.06	1.31	0.07	0.05	0.06	0.01	0.00	0.40	0.00	3.77
10	0.09	0.68	1.15	0.00	0.22	0.05	1.25	0.10	0.08	0.08	0.02	0.00	0.82	0.07	4.61
11	0.09	0.63	1.90	0.00	0.27	0.06	2.19	0.25	0.10	0.14	0.04	0.06	0.66	0.13	6.52
12	0.08	0.63	1.40	0.00	0.22	0.08	1.46	0.12	0.10	0.12	0.02	0.06	1.58	0.04	5.91
13	0.08	0.89	1.21	0.00	0.28	0.06	1.12	0.13	0.07	0.13	0.01	0.06	1.06	0.08	5.18
14	0.08	0.73	0.84	0.00	0.28	0.08	1.28	0.11	0.05	0.06	0.01	0.06	1.11	0.00	4.69
15	0.09	0.72	1.62	0.17	0.40	0.06	4.34	0.23	0.08	0.10	0.01	0.00	1.17	0.00	8.99
16	0.09	0.84	0.64	0.00	0.32	0.06	1.25	0.12	0.06	0.07	0.01	0.00	1.32	0.06	4.84
17	0.09	0.70	1.11	0.00	0.34	0.06	2.01	0.17	0.09	0.08	0.01	0.00	0.96	0.00	5.62
18	0.08	0.90	0.91	0.00	0.32	0.04	0.86	0.15	0.07	0.09	0.01	0.00	0.84	0.09	4.36
19	0.08	0.78	1.01	0.00	0.38	0.06	1.35	0.12	0.06	0.07	0.01	0.00	0.82	0.00	4.74
20	0.08	0.66	1.07	0.00	0.31	0.06	2.25	0.19	0.10	0.09	0.03	0.00	1.18	0.08	6.1
Mean	0.09	0.79	1.16	0.03	0.28	0.07	1.61	0.14	0.07	0.09	0.02	0.01	1.23	0.06	5.65
Media	0.09	0.77	1.13	0.00	0.27	0.06	1.38	0.13	0.07	0.08	0.01	0.00	1.04	0.05	5.08
DF (%)	100	100	100	15	100	100	100	100	100	100	100	20	100	60	

Table S10. Concentrations (ng/L) of PFASs in urine of the volunteers

	PFHxS	PFOS	PFHxA	PFHpA	PFOA	PFNA	PFDA	6:2 Cl-PFESA	ΣPFASs
1	3.28	9.15	9.98	2.58	23.73	2.60	1.12	8.74	61.18
2	2.88	10.03	5.62	4.02	13.41	3.79	0.50	3.32	43.57
3	4.66	5.97	8.49	1.89	25.33	1.54	0.51	3.48	51.87
4	3.19	7.35	8.26	1.66	35.93	1.68	0.59	6.00	64.66
5	2.52	12.16	8.17	2.15	28.66	1.65	0.50	1.90	57.71
6	3.41	4.29	7.73	1.55	10.92	1.33	0.42	7.54	37.19
7	2.92	7.36	11.55	2.42	25.00	1.21	0.48	1.64	52.58
8	3.05	4.84	17.89	1.01	19.52	1.86	0.55	1.96	50.68
9	5.40	6.15	13.65	4.02	21.19	1.13	0.45	2.07	54.06
10	3.86	7.62	11.49	2.51	18.53	1.22	0.62	4.85	50.7
11	2.52	13.20	6.38	10.84	36.59	3.85	0.85	1.82	76.05
12	2.64	8.94	9.64	1.94	21.23	1.84	0.87	1.55	48.65
13	1.41	10.35	6.68	3.32	16.85	1.61	0.60	2.61	43.43
14	2.48	5.99	7.32	0.60	18.63	1.74	0.44	3.06	40.26
15	1.97	10.41	4.43	2.18	49.91	2.80	0.67	2.72	75.09
16	1.61	3.83	7.41	1.36	19.99	1.63	0.50	2.65	38.98
17	1.42	7.15	8.28	0.86	28.36	2.58	0.72	3.24	52.61
18	1.26	6.40	9.01	2.00	13.81	1.73	0.62	3.30	38.13
19	1.44	6.67	7.43	1.20	23.74	1.34	0.50	3.19	45.51
20	1.88	6.57	3.94	1.45	39.12	3.27	0.79	2.88	59.9
Mean	2.69	7.72	8.67	2.48	24.52	2.02	0.62	3.43	
Median	2.58	7.25	8.22	1.97	22.48	1.70	0.57	2.97	
DF (%)	100	100	100	100	100	100	100	100	

Table S11. Spearman's correlation of PFASs in blood and urine samples

	p value	r
PFHxS	0.94	-0.017
PFOS	<0.001	0.97
PFHxA	<0.01	-0.64
PFHpA	0.17	0.31
PFOA	<0.001	0.94
PFNA	<0.001	0.83
PFDA	<0.001	0.98
6:2 Cl-PFESA	0.80	-0.059

Table S12. PFASs renal clearance rates (mL/day/kg)

	ΣPFHxS	ΣPFOS	PFHxA	PFHpA	ΣPFOA	PFNA	PFDA	6:2 Cl-PFESA
1	0.08	0.17	1.04	1.04	0.33	0.29	0.20	0.07
2	0.07	0.14	0.46	1.42	0.29	0.31	0.18	0.04
3	0.08	0.10	0.48	0.48	0.25	0.17	0.13	0.06
4	0.07	0.10	0.41	0.44	0.22	0.19	0.13	0.09
5	0.05	0.11	0.62	0.68	0.36	0.29	0.15	0.01
6	0.08	0.13	0.66	0.53	0.26	0.31	0.16	0.17
7	0.09	0.14	1.32	0.89	0.39	0.31	0.18	0.03
8	0.08	0.15	1.82	0.38	0.44	0.36	0.18	0.09
9	0.17	0.15	1.30	1.45	0.33	0.34	0.17	0.11
10	0.11	0.13	1.02	0.85	0.29	0.24	0.16	0.11
11	0.08	0.14	0.47	3.54	0.33	0.31	0.17	0.06
12	0.10	0.15	1.03	0.82	0.35	0.38	0.20	0.02
13	0.04	0.20	0.56	1.38	0.35	0.29	0.19	0.06
14	0.08	0.17	0.61	0.25	0.34	0.36	0.19	0.06
15	0.05	0.13	0.22	0.75	0.23	0.25	0.17	0.05
16	0.04	0.12	0.46	0.47	0.32	0.28	0.17	0.04
17	0.05	0.15	0.57	0.34	0.33	0.35	0.19	0.08
18	0.03	0.13	0.54	0.67	0.31	0.22	0.16	0.08
19	0.03	0.10	0.29	0.31	0.26	0.16	0.12	0.06
20	0.06	0.12	0.25	0.51	0.35	0.34	0.16	0.05
Mean	0.07	0.14	0.71	0.86	0.32	0.29	0.17	0.07

251 **Table S13. Data of renal clearance rates of PFASs reported by previous studies and this study**

Ref.		PFHxA	PFHpA	PFOA	PFNA	PFDA	PFUnA	PFHxS	PFOS	6:2 Cl-PFESA	
[9]	serum			0.03					0.02		general population
[10]	serum		0.61	0.53	0.20	0.08	0.06	0.03	0.04		
[11]	serum	7.97	3.26	0.15				6.47	0.02		background exposed reference persons
[12]	serum			0.08				0.04	0.01		occupational workers
this study	blood (serum)	0.71 (0.35)	0.86 (0.43)	0.32 (0.16)	0.29 (0.15)	0.17 (0.08)		0.07 (0.04)	0.14 (0.07)	0.07 (0.03)	general population
[13]	serum			0.11				0.03	0.04		
[14]	serum								0.01	0.003	general population
[14]	serum									0.0006	occupational workers

Table S14. Proportions (%) of PFOS, PFOA and PFHxS isomers in pooled food

	PFOS					PFOA					PFHxS				
	1m-	*(3+5)m-*	*4m-*	*iso-*	*n-*	*3m-*	*4m-*	*5m-*	*iso-*	*n-*	*1m-*	*2m-*	*3m-*	*iso-*	*n-*
1	0.8	17.4	1.2	2.0	78.7	2.3	1.7	0.3	6.8	89.0	1.6	1.7	1.0	3.0	92.6
2	0.1	17.4	4.6	8.1	69.7	1.9	1.1	0.7	5.8	90.5	2.3	2.0	0.8	2.7	92.2
3	0.1	2.2	0.6	0.7	96.4	2.6	1.3	4.1	7.8	84.3	1.9	0.8	1.8	2.7	92.8
4	0.9	19.5	0.5	15.8	63.3	2.4	0.9	2.6	7.2	86.9	3.2	1.0	2.6	1.8	91.5
5	0.3	11.7	5.4	2.1	80.5	2.6	0.7	1.1	7.8	87.8	1.8	0.9	1.2	2.1	94.0
6	0.1	25.5	2.5	0.2	71.7	1.5	0.8	1.3	4.4	91.9	2.2	2.4	2.2	1.7	91.5
7	0.1	13.4	4.2	0.7	81.7	3.3	1.4	2.7	9.9	82.8	1.6	4.1	1.1	1.6	91.6
8	0.0	1.0	0.5	0.3	98.2	3.2	0.6	0.1	9.7	86.3	4.5	1.9	2.5	2.5	88.5
9	0.1	4.5	4.7	1.1	89.5	1.7	0.6	1.3	5.0	91.4	3.0	1.6	1.4	1.6	92.5
10	0.6	11.6	0.9	4.2	82.6	2.8	0.8	3.2	8.5	84.8	2.1	0.4	2.5	6.0	89.0
11	1.7	16.9	5.0	2.2	74.2	2.7	0.7	0.4	8.0	88.2	3.0	1.2	1.0	9.3	85.6
12	0.4	6.1	0.3	0.6	92.6	3.0	1.8	3.5	9.1	82.5	2.0	0.3	0.9	7.2	89.6
13	0.1	3.8	0.6	0.6	94.9	2.8	1.4	3.6	8.3	83.9	5.3	4.5	1.2	17.5	71.5
14	0.2	12.4	0.9	4.0	82.4	6.7	1.1	4.7	20.0	67.5	4.3	2.0	9.3	5.4	78.9
15	0.1	16.0	0.5	4.7	78.7	4.0	3.1	5.5	12.1	75.3	1.1	3.7	2.0	5.1	88.1
16	0.2	28.8	0.9	4.1	66.0	7.2	1.9	7.1	21.6	62.1	1.8	1.9	6.5	5.8	83.9
17	0.6	6.6	1.3	1.6	90.0	3.7	1.3	2.5	11.1	81.4	4.0	0.3	6.5	5.4	83.8
18	0.6	6.4	3.2	0.9	89.0	2.9	0.8	2.0	8.8	85.4	2.7	2.0	1.3	5.7	88.3
19	0.4	10.4	0.6	1.0	87.5	6.5	1.8	3.5	19.6	68.6	3.6	1.9	1.7	1.5	91.3
20	0.6	8.7	0.5	3.5	86.8	4.6	1.3	4.9	13.7	75.5	2.5	0.3	1.3	1.1	94.8
Mean	0.4	12.0	1.9	2.9	82.7	3.4	1.3	2.8	10.3	82.3	2.7	1.8	2.5	4.5	88.6
SD	0.4	7.3	1.8	3.5	9.8	1.6	0.6	1.9	4.8	8.1	1.1	1.2	2.2	3.7	5.5

Table S15. Proportions (%) of PFOS, PFOA and PFHxS isomers in whole blood

	PFOS					PFOA					PFHxS				
	1m-	*(3+5)m-*	*4m-*	*iso-*	*n-*	*3m-*	*4m-*	*5m-*	*iso-*	*n-*	*1m-*	*2m-*	*3m-*	*iso-*	*n-*
1	2.1	15.8	4.9	4.4	72.9	1.6	0.8	0.5	3.0	94.1	0.3	1.1	1.4	4.8	92.5
2	1.2	28.5	15.8	4.5	49.9	1.5	0.7	0.0	2.8	95.0	0.1	0.6	1.6	2.1	95.6
3	0.3	19.3	6.6	4.5	69.4	1.2	0.4	0.4	2.1	95.9	0.5	0.5	2.2	2.8	94.0
4	2.3	21.2	7.7	4.5	64.3	1.5	0.6	0.1	2.7	95.1	0.8	1.0	3.0	7.1	88.2
5	0.7	41.4	17.3	2.8	37.8	0.8	0.6	0.6	1.5	96.5	0.0	0.5	1.4	4.2	93.9
6	2.0	19.1	6.8	4.8	67.3	1.1	0.6	1.0	1.9	95.4	0.0	0.6	6.1	5.2	88.1
7	1.9	15.8	4.2	2.5	75.5	1.3	1.0	1.7	2.4	93.6	1.2	0.4	3.8	4.8	89.8
8	2.4	21.5	6.9	2.3	66.8	0.5	0.5	0.6	1.0	97.4	1.7	0.1	4.4	7.2	86.6
9	1.4	14.0	5.3	5.7	73.7	0.5	0.7	0.0	0.9	97.8	1.6	0.1	7.8	8.2	82.2
10	1.5	15.7	5.3	2.9	74.6	0.7	0.6	0.6	1.3	96.8	1.1	0.4	3.7	5.1	89.7
11	1.0	14.3	5.6	1.9	77.3	0.3	0.5	0.2	0.5	98.6	0.5	0.7	3.3	7.1	88.4
12	1.7	15.4	6.5	2.2	74.2	0.4	0.3	0.1	0.7	98.5	0.5	0.6	1.9	3.0	94.0
13	1.7	14.6	6.9	2.8	74.1	0.2	0.0	0.9	0.4	98.4	0.3	0.1	2.7	4.1	92.9
14	1.6	20.9	8.9	4.8	63.9	0.2	0.2	0.0	0.3	99.3	0.1	0.1	3.1	5.1	91.6
15	1.1	16.9	7.4	3.8	70.9	0.2	0.9	0.0	0.3	98.6	1.5	0.4	2.7	4.4	90.9
16	2.0	18.0	8.2	2.8	69.0	0.3	0.6	0.1	0.5	98.5	0.2	0.6	3.5	5.3	90.3
17	1.9	17.1	6.6	5.8	68.6	0.1	0.9	0.1	0.3	98.5	0.7	0.1	0.0	5.1	94.1
18	1.4	15.5	6.2	5.4	71.5	0.1	0.9	0.2	0.2	98.7	1.1	0.0	7.4	9.3	82.2
19	5.1	21.6	8.2	1.7	63.3	0.1	0.1	0.2	0.1	99.5	0.6	0.5	0.0	2.5	96.4
20	1.8	16.5	5.7	3.9	72.2	1.0	0.7	3.0	1.8	93.6	0.6	0.1	3.8	5.6	89.9
Mean	1.8	19.2	7.5	3.7	67.9	0.7	0.6	0.5	1.2	97.0	0.7	0.4	3.2	5.1	90.6
SD	0.9	6.1	3.2	1.3	9.1	0.5	0.3	0.7	0.9	1.9	0.5	0.3	2.0	1.8	3.8

Table S16. Proportions (%) of PFOS, PFOA and PFHxS isomers in urine

	PFOS					PFOA					PFHxS				
	1m-	*(3+5)m-*	*4m-*	*iso-*	*n-*	*3m-*	*4m-*	*5m-*	*iso-*	*n-*	*1m-*	*2m-*	*3m-*	*iso-*	*n-*
1	1.5	21.1	8.7	13.6	55.2	0.0	2.2	2.4	2.6	92.8	0.0	2.1	2.6	4.9	90.4
2	1.3	21.3	7.6	13.1	56.6	0.0	1.7	2.8	2.2	93.3	0.0	1.6	3.4	5.2	89.9
3	1.2	21.3	7.2	12.4	58.0	0.0	1.9	2.1	2.5	93.6	0.0	1.3	3.4	6.2	89.1
4	1.4	20.7	9.2	13.0	55.7	0.0	1.2	1.6	1.8	95.5	0.0	0.9	2.5	3.6	93.0
5	1.4	19.1	7.8	11.9	59.7	0.0	1.1	1.8	1.9	95.1	0.0	1.5	2.8	6.5	89.2
6	1.3	22.6	7.3	14.3	54.6	0.0	1.1	2.2	1.9	94.8	0.0	1.4	2.6	5.5	90.5
7	1.1	20.0	7.3	10.9	60.6	0.0	0.6	1.3	1.6	96.4	0.0	1.7	2.9	6.5	88.9
8	1.1	20.2	9.9	13.3	55.5	0.0	0.5	1.2	2.1	96.1	0.0	1.6	2.7	6.6	89.0
9	1.2	18.8	10.0	13.1	56.9	0.0	0.9	1.7	2.1	95.3	0.0	1.0	3.0	4.6	91.4
10	1.4	18.0	9.5	13.1	58.1	0.0	1.4	1.9	2.2	94.5	0.0	1.3	3.0	4.2	91.5
11	1.7	17.0	8.9	13.7	58.7	0.0	0.9	1.6	2.0	95.5	0.0	1.8	3.5	5.6	89.1
12	1.3	16.6	10.0	12.7	59.3	0.0	1.4	2.0	2.6	94.1	0.0	1.1	3.8	5.4	89.6
13	1.1	22.3	7.0	11.0	58.6	0.0	1.1	1.7	2.0	95.2	0.0	2.1	3.0	6.4	88.5
14	1.2	22.0	8.0	10.4	58.3	0.0	0.9	1.6	1.9	95.5	0.0	1.9	3.4	5.7	89.1
15	1.2	21.7	7.9	9.0	60.2	0.0	1.4	2.0	1.7	94.9	0.0	1.7	3.4	5.5	89.5
16	1.2	22.6	7.1	7.8	61.3	0.0	1.2	1.5	1.8	95.5	0.0	1.6	2.8	5.3	90.3
17	1.4	22.6	7.9	9.6	58.6	0.0	1.3	2.0	1.9	94.8	0.0	1.9	3.5	5.4	89.3
18	1.3	23.2	7.6	9.5	58.5	0.0	1.3	1.8	2.3	94.6	0.0	1.7	3.8	6.6	87.8
19	1.4	22.1	7.1	8.8	60.5	0.0	1.3	1.5	1.6	95.6	0.0	1.7	4.7	5.6	88.0
20	1.2	22.0	7.3	8.9	60.7	0.0	1.1	1.8	2.0	95.1	0.0	1.9	4.6	6.0	87.5
Mean	1.3	20.8	8.2	11.5	58.3	0.0	1.2	1.8	2.0	94.9	0.0	1.6	3.3	5.6	89.6
SD	0.1	1.9	1.0	2.0	1.9	0.0	0.4	0.4	0.3	0.9	0.0	0.3	0.6	0.8	1.3

Table 17. *p* values of isomer compositions of EDI compared with PFHxS technical products and PFHxS impurity of PFOS industrial products using Kruskal-Wallis test.

	Isomers	PFHxS technical product	PFHxS impurity of PFOS industrial product
EDI	*1m*-PFHxS	**	**
	2m-PFHxS	0.36	*
	3m-PFHxS	0.64	0.07
	iso-PFHxS	0.06	**
	n-PFHxS	0.51	**

* represents $p<0.05$

** represents $p<0.01$

Table S18. Isomer-specific PFASs renal clearance (mL/day/kg)

	PFOS					PFOA					PFHxS				
	1m-	*(3+5)m-*	*4m-*	*iso-*	*n-*	*3m-*	*4m-*	*5m-*	*iso-*	*n-*	*1m-*	*2m-*	*3m-*	*iso-*	*n-*
1	0.12	0.23	0.30	0.52	0.13	N.A.	0.94	1.61	0.30	0.33	N.A.	0.15	0.14	0.08	0.07
2	0.15	0.10	0.07	0.40	0.16	N.A.	0.68	N.A.	0.23	0.28	N.A.	0.17	0.15	0.17	0.06
3	0.43	0.11	0.11	0.27	0.08	N.A.	1.13	1.19	0.30	0.25	N.A.	0.24	0.12	0.18	0.08
4	0.06	0.10	0.12	0.28	0.08	N.A.	0.48	2.74	0.15	0.22	N.A.	0.06	0.05	0.03	0.07
5	0.22	0.05	0.05	0.48	0.18	N.A.	0.73	1.01	0.48	0.36	N.A.	0.15	0.09	0.07	0.04
6	0.09	0.16	0.14	0.40	0.11	N.A.	0.52	0.55	0.26	0.26	N.A.	0.17	0.03	0.08	0.08
7	0.09	0.18	0.25	0.62	0.11	N.A.	0.25	0.30	0.27	0.41	N.A.	0.42	0.07	0.12	0.09
8	0.07	0.14	0.21	0.84	0.12	N.A.	0.50	0.86	0.95	0.43	N.A.	1.48	0.05	0.08	0.08
9	0.13	0.20	0.28	0.34	0.12	N.A.	0.44	13.80	0.78	0.33	N.A.	1.40	0.06	0.09	0.19
10	0.11	0.15	0.23	0.59	0.10	N.A.	0.63	0.86	0.50	0.28	N.A.	0.38	0.09	0.09	0.11
11	0.24	0.17	0.22	1.03	0.11	N.A.	0.59	2.22	1.48	0.32	N.A.	0.22	0.09	0.06	0.08
12	0.12	0.17	0.24	0.88	0.12	N.A.	1.53	13.80	1.25	0.33	N.A.	0.18	0.20	0.18	0.10
13	0.13	0.31	0.20	0.80	0.16	N.A.	N.A.	0.67	1.60	0.34	N.A.	1.50	0.04	0.06	0.04
14	0.13	0.18	0.15	0.36	0.15	N.A.	1.92	N.A.	1.95	0.33	N.A.	2.07	0.09	0.09	0.08
15	0.14	0.17	0.14	0.30	0.11	N.A.	0.37	N.A.	1.23	0.22	N.A.	0.23	0.07	0.07	0.05
16	0.07	0.15	0.10	0.34	0.11	N.A.	0.62	3.30	1.27	0.31	N.A.	0.10	0.03	0.04	0.04
17	0.11	0.20	0.18	0.25	0.13	N.A.	0.44	5.74	2.47	0.32	N.A.	1.09	N.A.	0.05	0.05
18	0.12	0.20	0.17	0.24	0.11	N.A.	0.48	3.39	4.54	0.30	N.A.	N.A.	0.01	0.02	0.03
19	0.03	0.10	0.08	0.51	0.09	N.A.	3.97	2.16	2.94	0.25	N.A.	0.10	N.A.	0.06	0.02
20	0.08	0.16	0.16	0.28	0.10	N.A.	0.56	0.21	0.39	0.35	N.A.	1.85	0.07	0.06	0.06
Mean	0.13	0.16	0.17	0.49	0.12	N.A.	0.88	3.20	1.17	0.31	N.A.	0.63	0.08	0.08	0.07

Table S19. Correlation of PFASs concentration between blood and EDI_{total}.

	p value	r
PFOS	0.42	0.19
PFOA	0.86	-0.04
PFHxS	0.51	0.15
6:2 Cl-PFESA	*	0.40

* represents the $p<0.05$.

Table S20: Temporal trends, 2009–2019, of PFAS concentrations in serum/blood of Tianjin residents

Year	Reference	matrix	PFHxS	PFOS	PFBA	PFHxA	PFHpA	PFOA	PFNA	PFDA	PFUdA	6:2 Cl-PFESA	8:2 Cl-PFESA
2009	[5]	blood	0.34	**12.5**		0.27	0.16	**0.49**	0.5	0.14	0.22		
2010	[15]	blood		**12.8**				**2.71**					
2012	[16]	blood	0.21	**2.37**				**2.54**	0.38	0.255	0.225		
2012	[17]	serum	0.67	**11.3(5.60*)**		0.39	0.12	**2.95(1.46*)**	0.93	1.17	0.52		
2015	[18]	serum	0.82	**14.5(7.40*)**	0.40	0.004	0.005	**8.73(4.36*)**	1.58	0.90	0.62	7.26	0.10
2016	[19]	serum	0.48	**11.6(5.60*)**		0.2	0.08	**12.3(6.20*)**	2.76	1.71	1.62	5.95	0.05
2017	[19]	serum	0.33	**14.2(7.10*)**	0.6	0.57	0.11	**14.8(7.40*)**	3.17	2.08	2.28	8.64	0.06
2019	this study	blood	0.79	**1.16**	0.03	0.28	0.07	**1.61**	0.14	0.07	0.09	1.23	0.06

*** the blood concentrations were derived from the serum concentrations based on the ratio of 1:2** [7]

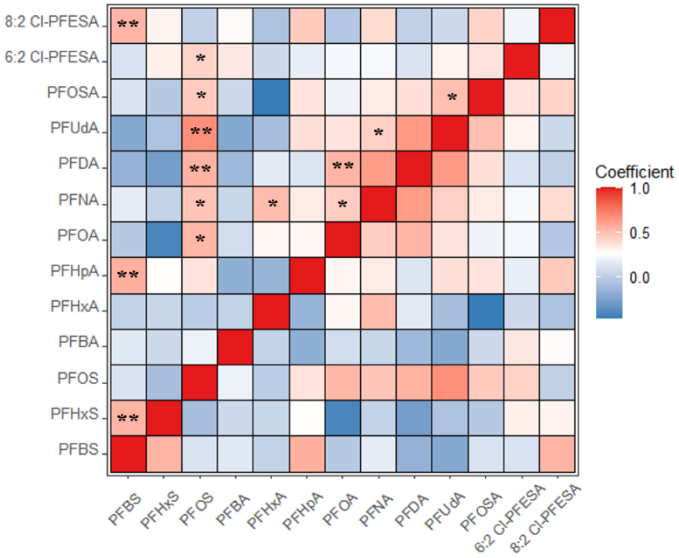


Figure S1. Correlation between PFASs in blood**.** * represents the $p<0.05$, ** represents the $p<0.01$.

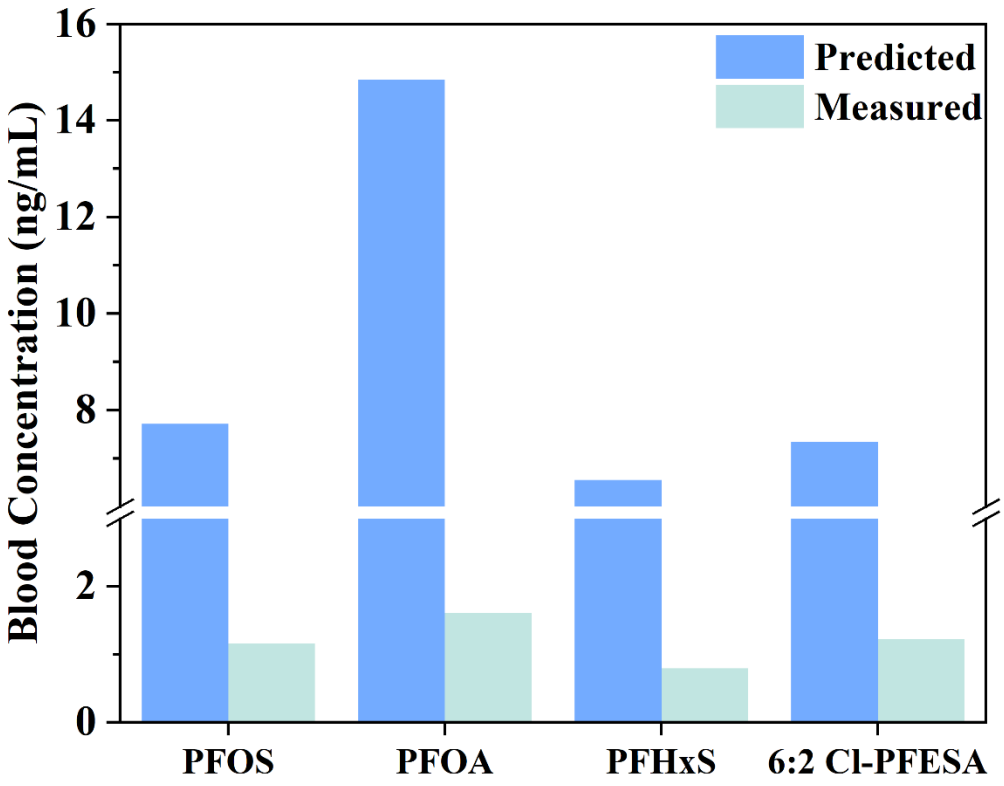


Figure S2. Comparison of predicted blood concentrations (F=100%) and measured concentrations.

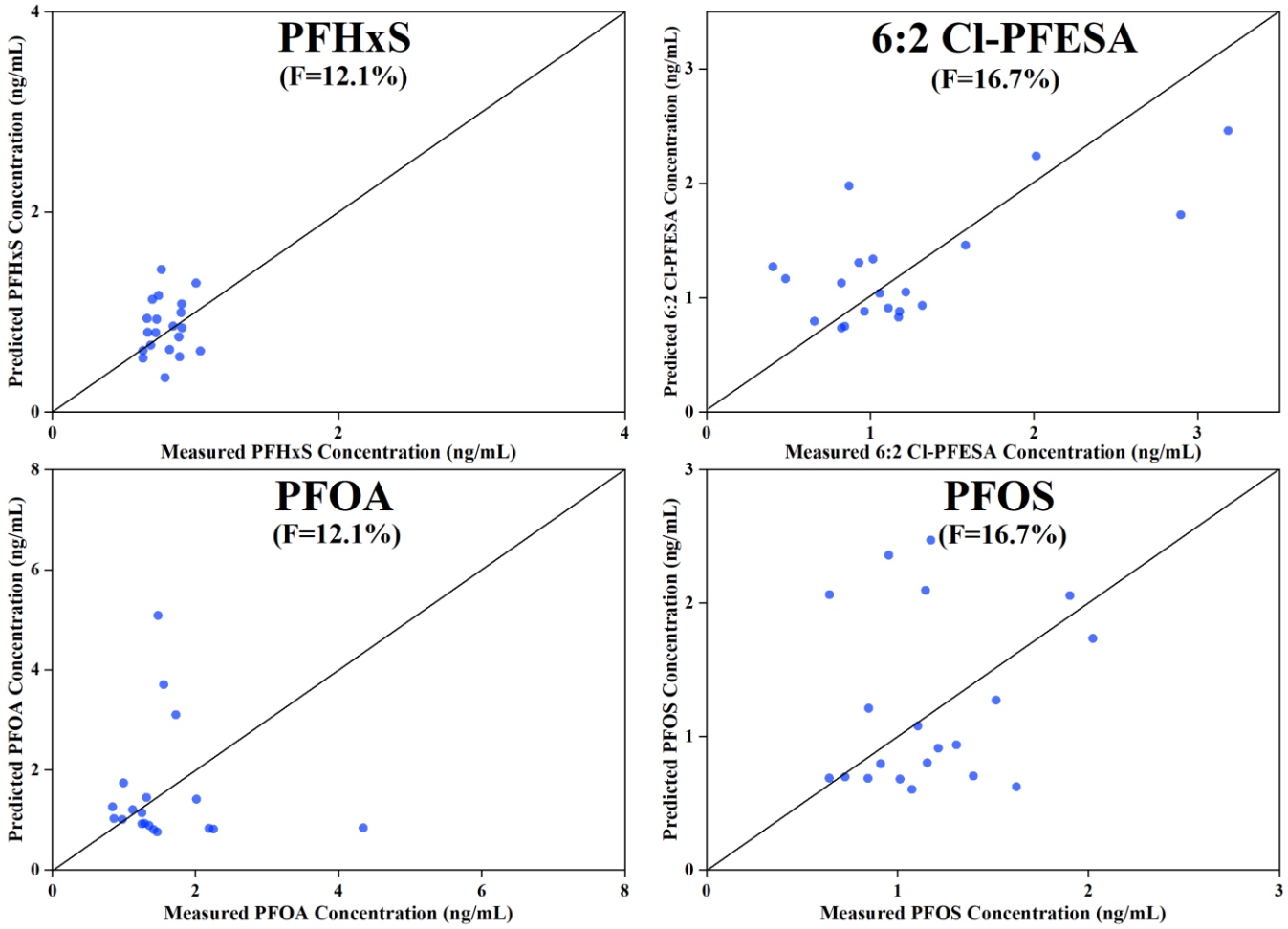


Figure S3. Scatter plots between measured and predicted blood concentrations (ng/mL) of PFHxS, 6:2 Cl-PFESA, PFOA and PFOS.

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